Harish G

Software Engineer | Platform Operations, Integration & Release Engineering

Bengaluru, India | +91 7892855850 | harishgreddy.work@gmail.com | LinkedIn | GitHub | harishg.com

Profile

Backend engineer who owns systems after they ship, not only up to the merge. On the platforms I have built I own the release path and the operational surface under it: multi-stage CI with separate staging and production gates, versioned schema migrations, containerised services, structured JSON logging correlated by request ID, latency instrumentation and a metrics endpoint, and deployment runbooks that name the failures the team actually hits. Java 21 with Spring Boot and Python with FastAPI, integrating upstream and downstream systems over REST, signed webhooks and bulk file transfer. Software Developer at CortexCraft.ai.

Core Skills

Deployment & Release Management: GitLab CI (lint, test, build, deploy with separate staging and production gates), Jenkins, GitHub Actions; Docker and Compose, multi-image builds; Helm charts with per-environment values; Terraform for managed cloud services; Alembic and Flyway versioned migrations; tagged releases with written hotfix and rollback procedures

Monitoring & Log Analysis: single-line structured JSON logging, `X-Request-ID` correlation end to end, per-request latency instrumentation, Prometheus `/metrics` endpoint, health checks, rate limiting, append-only audit trails

Integration: REST and webhook integration across upstream and downstream systems (Twilio, Plivo, WhatsApp Business API, Google Calendar, Jira, GitHub); HMAC signature verification with constant-time comparison; retry with configurable delay and attempt limits; bulk CSV and Excel file transfer; Celery and APScheduler job queues; Redis locks

Languages & Backend: Java 21, Python, JavaScript / TypeScript, SQL; Spring Boot 3.5 (Security, Data JPA, Validation, JJWT, Flyway), FastAPI, Node / Express; PostgreSQL, MongoDB, Redis

Defect Resolution & Stability: guarded state machines that reject invalid transitions outright, 92 pytest suites behind coverage gates, Playwright end-to-end tests, schema validation at the boundary, idempotent handlers

Documentation & Ways of Working: deployment runbooks with a symptom-to-fix section, release process and version-control SOPs, ADRs recorded before the code lands, one issue per branch, a changelog per release

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Authored 380 of 418 commits on the voice-agent SaaS platform, across the application, its integration layer and its release path.
- Sole author of the deploy tree. A four-stage GitLab pipeline runs lint, test, build and deploy with separate staging and production jobs; three container images build from it including a dedicated worker; a Helm chart carries per-environment values and Terraform provisions the managed services behind it.
- Treat schema and configuration change as releases: 27 Alembic migrations applied in order, per-tenant configuration held in the database rather than in code, and vendor credentials encrypted at rest.
- Maintain integration workflows against two telephony vendors and several partner APIs: signed webhook handlers verified with constant-time HMAC comparison, retry with configurable delay and attempt limits, and a generic HTTP executor so onboarding a new upstream system is a configuration change rather than a code change.
- Put the campaign lifecycle behind a guarded state machine with an explicit transition table, terminal states and a unit-tested rejection path returning HTTP 409, so a run cannot drift into an inconsistent state instead of failing loudly.
- Quality gate on every pipeline run: 92 pytest suites plus Playwright end-to-end tests.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

Delivered client projects end to end and then supported them: build, deployment, and the fixes, environment changes and upgrades afterwards. Direct client contact for scoping, release sign-off and issue triage.

Selected Projects

KisanVoice AI, multilingual survey platform *Python, FastAPI, MongoDB, Jenkins, Docker*

- 172 of 192 commits. Operations built in rather than added later: every request carries an `X-Request-ID` that is forwarded or generated and echoed back, every log line is a single JSON object carrying that ID, and each response is logged with its status and duration in milliseconds. A Prometheus instrumentator exposes `/metrics` with status codes grouped and health endpoints excluded.
- Ships with a deployment runbook covering prerequisites, environment setup, a verification step and a symptom-to-fix section for the failures the team actually hit, alongside a written release process with a hotfix flow and a rollback procedure. Jenkins builds it, Docker Compose runs it.
- Integration and bulk transfer: WhatsApp Business API flows, bulk CSV invite jobs and Excel round-trip, over an append-only audit log with TTL retention and ownership guards.

Lendly, lending platform backend *Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Docker*

- Spring Boot 3.5 on Java 21 with Spring Security and JJWT, Data JPA and Bean Validation, Flyway versioned migrations against PostgreSQL and H2 for tests, GitHub Actions CI and Docker Compose for local environments.

LeadCall AI, multi-tenant outbound calling *FastAPI, Celery, PostgreSQL, Redis, React*

- Asynchronous integration workload: Celery dial and callback workers behind Redis dial locks, per-tenant telephony credentials encrypted at rest, Excel lead import, and an append-only audit log of every operator action.

Education & Ways of Working

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore. One issue per branch, decisions recorded as ADRs before the code lands, tests as the merge gate.